

Automation with Gen AI

All Capstone Projects — Quick Reference

36 Projects · Set A · Set B · Set C

Each project is described in three lines: what to build, what tech to use, and what the expected output is. Choose one project from any set. Read the brief, understand what is expected, then build.

Set A

Track 1 · AI Model Focused

01. Prompt battle arena

Build: Build a Streamlit app that sends the same prompt to Claude, GPT-4o, and Gemini simultaneously, uses a fourth AI call to score each response against a user-defined rubric, and displays the scored comparison.

Use: Claude API + OpenAI API + Gemini API + Streamlit

Expect: Scored side-by-side comparison with winner declared per criterion and a downloadable PDF battle report.

02. Contract risk scanner

Build: Build a Streamlit app that accepts a PDF or pasted contract text, identifies risky clauses, missing protections, and liability exposure, and rates each finding as High, Medium, or Low severity.

Use: Claude API + Streamlit + pypdf

Expect: Structured risk report with severity ratings displayed in the app and downloadable as a formatted PDF.

03. Cold outreach personaliser

Build: Build a Streamlit app that takes a prospect profile and generates a personalised outreach email with subject line and follow-up. Batch mode accepts a CSV of prospects and generates emails for all.

Use: Claude API + Streamlit

Expect: Individual email output with tone selector, plus batch CSV download with personalised emails for every row.

04. Study guide generator

Build: Build a Streamlit app that takes a lecture PDF or syllabus text and generates summary notes, 20 MCQs with answers, 5 short-answer questions, a glossary, and a study order based on selected difficulty level.

Use: Claude API + Streamlit + pypdf

Expect: Complete downloadable study pack as PDF and MCQ export as CSV compatible with Anki flashcard import.

05. Code reviewer and explainer

Build: Build a Streamlit app where a user pastes code in any language, selects review depth, and receives a plain-English explanation, bug list with line references, refactored version, and complexity assessment.

Use: Claude API + Streamlit

Expect: Side-by-side original vs refactored code view in the app and a downloadable markdown review file.

Track 2 · Automation Focused

06. Job application autopilot

Build: Build an n8n workflow triggered by Google Form where a user submits a CV and job description. AI scores fit, generates a personalised cover letter, emails it to the applicant, logs to Sheets, and sets a 7-day follow-up calendar reminder.

Use: n8n + Claude API + Google Forms + Google Sheets + Gmail + Google Calendar

Expect: Automated cover letter emailed to applicant, fit score logged to Sheets, and calendar reminder created with no manual steps.

07. Social media content pipeline

Build: Build an n8n workflow where a content idea submitted via Google Form triggers AI generation of a LinkedIn post, Twitter thread, and Instagram caption. Drafts posted to Slack for approval. Approval emoji logs content to a publishing calendar Sheet.

Use: n8n + Claude API + Google Forms + Google Sheets + Slack

Expect: Three platform-specific drafts in Slack within 60 seconds of form submission, with approval-triggered logging to a publishing calendar.

08. Customer feedback intelligence

Build: Build an n8n workflow that triggers on incoming email, classifies it into bug/feature/complaint/praise, extracts product and severity, routes to correct Slack channel, logs to Sheets, and sends a weekly trend summary email every Monday.

Use: n8n + Claude API + Gmail + Google Sheets + Slack

Expect: Real-time Slack routing per category, structured Sheet log, and automated Monday morning digest email to the product team.

09. Interview question generator

Build: Build an n8n workflow where a hiring manager submits a job description via Google Form. AI generates 20 interview questions across technical, behavioural, and situational categories with scoring rubrics, delivered as PDF to the hiring manager's email.

Use: n8n + Claude API + Google Forms + Google Sheets + Gmail

Expect: Complete interview kit PDF emailed automatically within 2 minutes of form submission, with seniority level visibly affecting question depth.

Track 3 · Combined

10. Research assistant with memory

Build: Build a system where n8n ingests URLs via webhook, scrapes and embeds content into ChromaDB, and a Streamlit app lets users query across all sources with Claude answering with citations showing which URL each fact came from.

Use: n8n + Claude API + OpenAI Embeddings + ChromaDB + Streamlit

Expect: Multi-source Q&A; with citations, persistent knowledge base across sessions, and downloadable PDF of the full Q&A; session.

11. Meeting intelligence dashboard

Build: Build a system where n8n receives meeting transcripts via webhook, AI extracts action items with owners and deadlines, decisions, and open questions. A Streamlit dashboard shows all meetings with owner filtering and overdue highlighting.

Use: n8n + Claude API + Streamlit

Expect: Multi-meeting dashboard with owner filter, overdue items highlighted in red, and CSV export of all action items across all meetings.

12. Competitor intelligence tracker

Build: Build a system where n8n fetches competitor content weekly, AI extracts announcements and positioning shifts, and a Streamlit dashboard shows a tagged timeline per competitor with a Monday morning email digest sent automatically.

Use: n8n + Gemini API + Streamlit + Gmail

Expect: Automated weekly digest email, filterable Streamlit timeline, and exportable weekly report as PDF with all sources cited.

Set B

Track 1 • AI Model Focused

01. Document Q&A;

Build: Build a Streamlit app where a user uploads a PDF and asks questions. The app chunks the document, retrieves relevant sections, and Claude answers with source references. If the answer is not in the document, Claude says so.

Use: Claude API + Streamlit + pypdf

Expect: Grounded answers with page or chunk references, downloadable Q&A; session as PDF, graceful handling of unsupported queries.

02. Resume screener

Build: Build a Streamlit app where a user uploads a job description and multiple resumes. AI scores each resume against the JD on at least 4 criteria and ranks all candidates from highest to lowest fit.

Use: Claude API + Streamlit + pypdf

Expect: Ranked candidate list with score breakdown per criterion, downloadable as CSV with all scores and reasoning.

03. Invoice processor

Build: Build a Streamlit app that accepts invoice PDFs or images and extracts vendor name, invoice number, date, all line items, subtotal, tax, and total into a structured table.

Use: GPT-4o Vision API + Streamlit

Expect: Structured data table in app with missing fields marked clearly, downloadable as Excel with proper column headers.

04. Multi-model comparator

Build: Build a Streamlit app that sends any prompt to Claude, GPT-4o, and Gemini simultaneously and displays all three responses side by side with latency shown per model. User can rate each response.

Use: Claude API + OpenAI API + Gemini API + Streamlit

Expect: Side-by-side responses with latency, user star ratings, and downloadable comparison PDF including prompt, responses, and ratings.

05. Content repurposer

Build: Build a Streamlit app where a user pastes an article and selects a target audience. AI generates a LinkedIn post, Twitter thread, and email newsletter version from the same content.

Use: Claude API + Streamlit

Expect: Three distinct platform outputs with LinkedIn character count shown, downloadable as a single formatted document.

06. Meeting notes processor

Build: Build a Streamlit app where a user pastes a meeting transcript. AI extracts action items with owners and deadlines, key decisions, open questions, and a 3-line executive summary.

Use: Claude API + Streamlit

Expect: Four structured output sections in the app, full report downloadable as PDF, action items exportable as CSV.

07. Structured data extractor

Build: Build a Streamlit app where a user defines extraction fields and pastes unstructured text. AI extracts exactly those fields. Batch mode accepts a CSV and extracts from every row.

Use: Claude API + Streamlit

Expect: Single and batch extraction modes, JSON display in app, batch results downloadable as CSV with one column per extracted field.

Track 2 · Automation Focused

08. Lead qualification pipeline

Build: Build an n8n workflow where a Google Form submission triggers AI lead scoring (0-100) with hot/warm/cold classification, writes to Sheets, and sends a formatted Slack alert to the sales channel.

Use: n8n + Claude API + Google Forms + Google Sheets + Slack

Expect: Automated Sheets entry and Slack alert within 60 seconds of submission, with different message format per lead classification.

09. Email classifier and router

Build: Build an n8n workflow triggered by incoming email. AI classifies into support/sales/complaint/internal with urgency level, routes summary to the correct Slack channel, and logs to Sheets.

Use: n8n + Claude API + Gmail + Google Sheets + Slack

Expect: Correct Slack routing per category with urgency visible in message, structured Sheet log, ambiguous emails routed to a review channel.

10. Content approval workflow

Build: Build an n8n workflow where a content brief submitted via Google Form triggers AI drafting, posts the draft to Slack for review, and approval or rejection emoji updates a Google Sheets publishing calendar.

Use: n8n + Claude API + Google Forms + Google Sheets + Slack

Expect: AI draft in Slack within 2 minutes of submission, emoji-triggered status update in Sheets, word count shown in Slack message.

11. Job application tracker

Build: Build an n8n workflow where a job applicant submits CV and JD via Google Form. AI extracts a structured candidate profile and fit score. Hiring manager and applicant both receive automated emails.

Use: n8n + Claude API + Google Forms + Google Sheets + Gmail

Expect: Two automated emails sent within 2 minutes, structured Sheet entry with extracted profile and fit score, failed AI calls handled with fallback notification.

Track 3 • Combined

12. RAG chatbot

Build: Build a system where n8n ingests documents via webhook into ChromaDB and a Streamlit chat interface lets users ask questions. Claude retrieves relevant chunks and answers with source citations.

Use: n8n + Claude API + OpenAI Embeddings + ChromaDB + Streamlit

Expect: Chat interface with persistent message history, every answer citing its source chunk, and full chat history exportable as text file.

13. Customer support dashboard

Build: Build a system where n8n receives support tickets via webhook, AI classifies by category and urgency and drafts a response. A Streamlit dashboard shows the live ticket queue with filter and resolve functionality.

Use: n8n + Claude API + Streamlit

Expect: Live ticket queue with category and urgency filters, AI draft visible per ticket, mark as resolved button, data persisted across app restarts.

14. Research aggregator

Build: Build a system where n8n fetches content from configured URLs, AI summarises each source and extracts key points, and a Streamlit dashboard displays a consolidated research report with all sources.

Use: n8n + Claude API + Streamlit

Expect: Consolidated report with per-source summary, key points, and credibility note, new URLs addable without losing previous research, full report exportable as PDF.

Set C

Voice & Audio

01. Mock interview coach

Build: Build a Streamlit app where a user selects an interview type, receives a question, submits their answer, and gets AI-evaluated feedback on clarity, relevance, and confidence — delivered as spoken audio via ElevenLabs.

Use: Claude API + ElevenLabs API + Streamlit

Expect: At least 10 questions per interview type, written and spoken feedback per answer, downloadable session report with scores.

02. Multilingual document narrator

Build: Build a Streamlit app where a user uploads a PDF, selects a language from at least 8 options, and the AI summarises the document and converts the summary to spoken audio in the selected language.

Use: [Claude API + ElevenLabs API + Streamlit + pypdf](#)

Expect: Accurate summary and downloadable MP3 audio file in the chosen language, graceful fallback for unsupported languages.

Multimodal

03. Product listing generator

Build: Build a Streamlit app where a user uploads product photos and selects a target platform style. The AI generates a complete e-commerce listing: title, 5 bullet features, full description, and keywords.

Use: [GPT-4o Vision API + Streamlit](#)

Expect: All four listing components generated per photo, platform style visibly affecting output, one-click copy or download of the complete listing.

04. Whiteboard to documentation converter

Build: Build a Streamlit app where a user uploads a whiteboard photo. The AI converts the content into structured documentation with headings, bullets, and plain-English descriptions of any diagrams.

Use: [GPT-4o Vision API + Streamlit](#)

Expect: Structured output separating text from diagram content, downloadable as PDF, with a confidence note where image is unclear.

05. Food nutrition analyser

Build: Build a Streamlit app where a user uploads a meal photo or food label. The AI estimates nutritional content, flags common allergens, and suggests a healthier alternative for the main ingredient.

Use: [GPT-4o Vision API + Streamlit](#)

Expect: Nutrition panel in the app, allergens highlighted visually, healthier alternative with brief reason, and a disclaimer that output is an estimate.

Agent-Style

06. Personalised news briefing agent

Build: Build a system where a user configures their interests and sources. An n8n cron job runs daily, fetches content, AI filters relevant articles and generates summaries, and the briefing is delivered by email every morning automatically.

Use: [n8n + Claude API + Gmail](#)

Expect: Fully automated daily email with at least 5 relevant article summaries, each with headline, 2-line summary, and source link — delivered at a configured time.

07. GitHub repo explainer

Build: Build a Streamlit app where a user pastes a public GitHub URL. The app fetches the README, file structure, and key files via the GitHub API. Claude generates a plain-English explanation covering what it does, how it is built, and how to get started.

Use: Claude API + GitHub API + Streamlit

Expect: Four-section structured explanation generated from live repo data, downloadable as markdown.

Real-World Tools

08. Salary negotiation coach

Build: Build a Streamlit app where a user pastes a job offer letter and their target salary. AI identifies negotiation levers and generates a professional counter-offer email with a negotiation strategy.

Use: Claude API + Streamlit

Expect: Structured negotiation brief with identified levers, recommended counter-offer range with reasoning, professional email draft, tone selector, and downloadable PDF.

09. College application essay reviewer

Build: Build a Streamlit app where a student pastes their personal statement and the prompt it answers. AI evaluates on 4 criteria, gives paragraph-level feedback, and rewrites the weakest paragraph as an example.

Use: Claude API + Streamlit

Expect: Score out of 100 with breakdown per criterion, specific paragraph comments, rewritten example paragraph, and downloadable marked-up review PDF.

10. API documentation generator

Build: Build a Streamlit app where a developer pastes a Python script. AI generates complete API documentation for every function: description, parameters, return values, usage examples, and edge case notes.

Use: Claude API + Streamlit

Expect: Complete documentation for every function detected, formatted as standard API docs, downloadable as markdown and rendered in the app.
